Relation-Conditioned Modality Gating for Student Performance Prediction in Heterogeneous Graphs

Anonymous Author(s)
Anonymous Institution

ABSTRACT

Predicting student academic performance requires integrating structured records, textual content, and behavioral logs — modalities that carry different signals depending on the relational context in which they are used. Existing heterogeneous graph neural networks (HGNNs) collapse these modalities into a single node embedding and apply uniform message passing regardless of edge type, discarding relation-specific modal relevance. We propose a relation-conditioned modality-gated heterogeneous graph neural network (**MG-HGNN**) in which each node maintains separate modality embeddings and a learnable, per-edge-type softmax gate $\boldsymbol{\alpha}^{(r)}$ controls how much each modality contributes to message passing along relation r. Evaluated on three datasets (OULAD, ASSISTments 2009, ASSISTments 2015), MG-HGNN consistently outperforms the identical HGTConv backbone without gating by +0.009–+0.042 AUC-ROC, directly isolating the contribution of relation-conditioned modal gating. On OULAD, MG-HGNN achieves RMSE 1.5518 ± 0.0913, AUC-ROC 0.8392 ± 0.0362, and Engagement-F1 0.5048 ± 0.0167 across 5-fold cross-validation. Ablation and gate-weight analyses confirm that edge-conditioned modal gating is the key driver of performance gains across all three datasets.

CCS CONCEPTS

• **Computing methodologies → Neural networks**; *Artificial intelligence.*

KEYWORDS

heterogeneous graph neural networks, student performance prediction, multi-modal learning, modality gating, educational data mining, learning analytics

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1 INTRODUCTION

Student academic performance prediction is a cornerstone task in educational data mining [31] and learning analytics [30], enabling institutions to identify at-risk learners early and deliver timely, targeted interventions. The stakes are high: in large-scale online education, dropout rates frequently exceed 50% [8], and even in campus settings, late detection of struggling students can have lasting consequences on retention and degree completion.

Early approaches to this problem treated students as independent instances, applying classical machine learning methods [25, 26] — logistic regression, support vector machines, and gradient-boosted trees — to flat per-student feature vectors assembled from demographic records, prior grades, and assignment scores. Complementary work in knowledge tracing [22–24] models student mastery at the exercise level using sequential models, but treats each student in isolation without relational graph structure. While effective on structured data, these methods discard the rich relational structure that characterises real educational ecosystems: students interact with courses, collaborate with peers, consume learning resources, and are guided by instructors. Graph neural networks (GNNs) have emerged as the natural framework for capturing these relations, with recent work demonstrating that explicit relational modelling consistently outperforms flat-feature baselines [7, 9, 10].

Despite this progress, a fundamental limitation persists in all existing heterogeneous GNN (HGNN) approaches: they collapse all available data modalities — structured academic records, textual content, and behavioural interaction logs — into a *single, unified node embedding* before message passing begins. This design ignores a key educational insight: different relation types in the student graph are informationally asymmetric. A student's clickstream behaviour on a learning resource is most predictive when propagating information through an *accessed* edge, but the same student's discussion posts and assignment text are more relevant when propagating through a *collaborated-with* edge. Multi-modal learning [42] has demonstrated that distinct data sources carry complementary signals; yet uniform modality fusion across all edge types conflates them and limits model expressiveness.

This paper proposes MG-HGNN, a *Modality-Gated Heterogeneous Graph Neural Network* that addresses this gap with a single, lightweight architectural addition. Each node maintains three *separate* modality embeddings, produced by a multi-layer perceptron (structured), a frozen BERT encoder (textual), and a bidirectional LSTM (behavioural). The core innovation is a per-edge-type *modality gate*: for each relation type $r \in \mathcal{R}$, a dedicated linear layer W_r learns a softmax weight vector $\boldsymbol{\alpha}^{(r)} = [\alpha_s^{(r)}, \alpha_t^{(r)}, \alpha_b^{(r)}]$ that dynamically controls how much each modality contributes to message passing along that specific relation. The gate adds only 4,620 parameters (0.53% of total trainable parameters) and is learned end-to-end with no manual specification of modal preferences.

Contributions. This paper makes four contributions:

(1) We identify and formalise the *modal asymmetry problem* in educational heterogeneous graphs: the empirical observation

that different relation types carry different modal signals, which uniform HGNN fusion fails to exploit.

(2) We propose MG-HGNN, which introduces the first *edge-type-conditioned* modality gate in heterogeneous graph neural networks. Unlike concurrent work on multi-modal HGNNs [34], which weights modalities globally per node, MG-HGNN learns a separate softmax gate W_r for each edge relation type r, allowing modal preferences to vary by relational context. This is a strictly finer-grained conditioning than any existing approach.

(3) We evaluate MG-HGNN on three public datasets (OULAD, ASSISTments 2009 [35], ASSISTments 2015 [36]), demonstrating consistent gains of +0.009–+0.042 AUC-ROC over the identical HGTConv backbone without gating across all datasets and cohort sizes — directly quantifying the value of relation-conditioned modal fusion.

(4) We provide interpretable gate-weight analysis showing the model autonomously discovers educationally meaningful modal preferences without supervision, and an ablation study confirming each component's contribution.

2 RELATED WORK

Educational data mining [27] and learning analytics have studied student performance prediction for over a decade, and the field has evolved substantially, progressing from classical machine learning on independent per-student feature vectors toward relational deep learning methods [15] that explicitly encode the complex network structure of educational ecosystems.

2.1 Graph Neural Networks for Student Performance Prediction

Early graph-based approaches showed that modelling student similarity as an explicit graph yields systematic gains over flat-feature baselines. Li et al. [9] proposed Study-GNN, a multi-topology GNN (MTGNN) that constructs multiple student similarity graphs using distinct distance metrics and fuses them via attention, casting performance prediction as semi-supervised node classification. Study-GNN outperformed SVMs, logistic regression, shallow neural networks, and vanilla GCN [11], validating relational modelling for education.

Huang and Zeng [7] introduced APP-DGNN, a dual graph neural network that constructs two complementary graphs: one from interaction logs capturing engagement dynamics, and one from student attributes such as demographics and prior grades. Fusing local and global representations, APP-DGNN achieved approximately 84% accuracy on pass/fail and 90% on pass/withdraw tasks on a public MOOC dataset. This dual-graph paradigm establishes a key principle: different information sources benefit from distinct graph topologies rather than a single aggregated representation.

2.2 Temporal and Dynamic Graph Models

Static graph methods cannot capture behavioral evolution across time. Huang and Chen [8] addressed this with APP-TGN, a temporal graph network encoding student activity logs as a dynamic graph. Combining low-pass and high-pass spectral filters with multi-head attention, APP-TGN significantly improves both overall accuracy and early at-risk detection in MOOC settings. Zhang et al. [33] embedded GNN layers inside a GNN-Transformer-InceptionNet hybrid for multi-label performance prediction, reporting up to 98.5% accuracy on a single-institution dataset.

Despite these advances, long-range temporal dependencies remain under-exploited in several dual and multi-graph designs [6–8], motivating continued work on temporally expressive architectures. MG-HGNN addresses a complementary and previously unexplored axis: *modal selectivity* during heterogeneous message passing.

2.3 Hypergraph and Higher-Order Models

Yang et al. [10] proposed FD-HGNN, a framelet-based dual hypergraph GNN using low- and high-pass hypergraphs to capture higher-order, multi-way student relationships. Evaluated on four datasets, FD-HGNN outperforms both traditional ML and standard GNNs. While this line of work captures structural complexity beyond pairwise edges, hypergraph constructions do not address the problem of multi-modal feature heterogeneity *within* each node — the gap targeted by MG-HGNN.

2.4 Multi-source and Heterogeneous Data Fusion

Several works highlight the need to handle sparse, multi-source data — behavioral logs, demographics, emotional indicators, and prior grades — more effectively than flat concatenation allows [6–10, 33]. Balachandar and Venkatesh [6] proposed MSPP, which applies advanced GNN layers on heterogeneous academic data, achieving approximately 76% accuracy on complex multi-class tasks. Across the literature, reported accuracy spans from ~76% on complex multi-class settings [6] to 84–90% on binary outcomes [7] and up to 98–99% on richer, single-institution benchmarks [33]. Shared limitations include high computational cost on large cohorts and limited cross-institutional generalizability [6–10, 33].

2.5 Gating and Modality Weighting in Graph Neural Networks

Gating mechanisms have been applied in GNNs primarily for *temporal* or *structural* selectivity, not multi-modal feature fusion. Gated Graph Neural Networks (GGNN) [32] introduced GRU-style recurrent gates to control information propagation depth across multiple message-passing steps; their gate conditions on node hidden state rather than input modality. In attention-based GNNs [3, 5], learned edge attention weights determine *which neighbours* contribute to aggregation, but all neighbours share the same fused node feature as their message — modality selection still does not occur. Early heterogeneous embedding methods such as metapath2vec [39] use meta-path guided random walks to learn node representations; MAGNN [40] aggregates messages across multiple meta-paths; and HetSANN [41] projects heterogeneous nodes into unified semantic spaces — yet none of these approaches perform edge-type-conditioned modality weighting.

Several multi-modal GNN papers in the vision–language domain introduce cross-modal attention to fuse image and text embeddings [10], but these fuse modalities at the *node level* before message passing, applying the same fused representation across all edge types. The heterogeneous nature of educational graphs — where the same student node participates in semantically distinct relation types simultaneously — demands *edge-conditioned* modality fusion: the optimal modality mixture for propagating information through an *accessed* edge (where clickstream behaviour dominates) differs fundamentally from the optimal mixture for an *enrolled_in* edge (where demographics and academic background dominate).

MG-HGNN is, to our knowledge, the first GNN architecture to condition multi-modal feature fusion on the *type of edge being traversed*, rather than applying a fixed or globally-learned fusion strategy uniformly across all relations. This positions MG-HGNN as complementary to existing attention mechanisms in HGNNs: HGT's attention selects *which node* matters; MG-HGNN's gate selects *which modality* matters, and the two can be combined.

2.6 Multi-modal Fusion in Graph Models

Recent work has explored fusing multiple data modalities within graph-structured models. MMGCN [19] uses modality-specific graphs for micro-video recommendation, maintaining separate GCN branches per modality and fusing their outputs at the prediction layer. Zadeh et al. [20] propose a memory fusion network for multi-modal sentiment analysis that explicitly models cross-modal interactions in a shared memory space. Cross-modal attention mechanisms [21] have demonstrated that different modalities benefit from relation-specific weighting during fusion: in ViLBERT, image regions and text tokens attend to each other through a co-attentional transformer rather than a single merged representation. For a comprehensive treatment of multimodal fusion strategies see [44].

However, none of these approaches extend to heterogeneous graphs with typed edges. MMGCN fuses at the output layer after modality-homogeneous message passing; memory fusion networks operate on sequences, not graph structures; and cross-modal attention in vision–language models conditions on token content, not on the *type of graph edge being traversed*. The precise gap MG-HGNN addresses is therefore orthogonal to this body of work: we ask not "how should image and text tokens attend to each other?" but "for this specific relation type in the student graph, which modality should dominate the message?"

Most recently, Bachiri et al. [34] proposed MM-HGNN, which introduces a Modality Transferability Function (MTF) to quantify cross-modal heterogeneity by measuring the performance gap when transferring a model trained on one modality to another. Their modality-level attention then prioritises modalities with lower transferability (higher uniqueness) during node embedding. While MM-HGNN represents a significant advance in multi-modal heterogeneous graph learning, it shares a key limitation with all prior work: the modality attention weights are computed globally per node, independent of which edge relation type is being traversed. A student's behavioral clickstream carries different relevance when propagating through a resource-access edge versus a peer-collaboration edge, yet MM-HGNN — like HAN [4] with its meta-path [38] based attention aggregation, HGT, and R-GCN [13] — applies the same modal weighting regardless of relational context. MG-HGNN addresses this gap directly by conditioning the modality gate on the edge relation type, learning a separate weight matrix W_r per relation $r \in \mathcal{R}$. This is the first mechanism of its kind in either general or educational HGNNs.

Table 1: Notation used throughout the paper.

Symbol	Description
$\mathcal{G}$	Heterogeneous educational graph
$\mathcal{V}$, $\mathcal{E}$	Node and edge sets
$\mathcal{T}_v$, $\mathcal{T}_e$	Node and edge type sets
$\tau(v)$	Type function for node v
N_s, N_c, N_r	# students, courses, resources
d	Shared embedding dimension (128)
d_s, d_t, d_b	Input dims: struct (64), text (768), behav (32)
T	Behavioural sequence length (50)
$\mathbf{X}_s$, $\mathbf{X}_t$, $\mathbf{X}_b$	Raw modality feature matrices
$\mathbf{h}_s^v$, $\mathbf{h}_t^v$, $\mathbf{h}_b^v$	Modality embeddings for node v
$W_r^g \in \mathbb{R}^{3\times 3d}$	Gate matrix for relation r
$\mathbf{b}_r \in \mathbb{R}^3$	Gate bias for relation r
$\boldsymbol{\alpha}^{(r)}(u)$	Softmax gate weights, relation r, source u
$\tilde{\mathbf{h}}_u^{(r)}$	Gated embedding of u for relation r
$\mathbf{h}'_v$	Final student embedding post-HGTConv
λ_g, λ_d, λ_e	Multi-task loss weights
$\lvert\mathcal{T}_e\rvert$	Number of relation types (4)

2.7 Gap Addressed by This Work

Despite the breadth of existing approaches, one assumption has remained unexamined: that all edge relation types should draw on the same mixture of input modalities when aggregating neighborhood information. Heterogeneous graph fusion for incomplete multimodal data [45] has been studied in recommendation settings, but without edge-type conditioning of modal weights. Typed-edge models such as R-GCN [13] use relation-specific weight matrices but perform no modality selection; and despite theoretical analysis of GNN expressiveness [14], no existing HGNN framework conditions its feature fusion strategy on the edge type being traversed. MG-HGNN fills this gap by maintaining separate per-modality embeddings at each node and learning a distinct, softmax-normalized modality gate $\boldsymbol{\alpha}^{(r)}$ for each relation $r \in \mathcal{R}$, allowing the model to discover which information sources are most relevant for each relational context without manual specification.

3 PROBLEM DEFINITION

Heterogeneous educational graph. We model the student academic ecosystem as a heterogeneous information network [37], formalized as a heterogeneous graph $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathcal{T}_v, \mathcal{T}_e)$, where $\mathcal{V}$ is the set of nodes, $\mathcal{E}$ the set of edges, $\mathcal{T}_v$ the set of node types, and $\mathcal{T}_e$ the set of edge relation types. Each node $v \in \mathcal{V}$ has a type $\tau(v) \in \mathcal{T}_v$, and each edge $(u, v, r) \in \mathcal{E}$ has a relation type $r \in \mathcal{T}_e$.

Node types. $\mathcal{T}_v = \{$*student*, *course*, *resource*, *instructor*$\}$. Student nodes carry three modality feature matrices: structured $\mathbf{X}_s \in \mathbb{R}^{N_s\times d_s}$, textual $\mathbf{X}_t \in \mathbb{R}^{N_s\times d_t}$, and behavioural $\mathbf{X}_b \in \mathbb{R}^{N_s\times T\times d_b}$, where T is the sequence length. Course and resource nodes carry textual features.

Edge relation types. $\mathcal{T}_e = \{$*enrolled_in*, *collaborated_with*, *accessed*, *submitted_to*$\}$, connecting student nodes to course, peer, resource, and assessment nodes respectively.

Prediction tasks. Given $\mathcal{G}$ and partial labels on student nodes, we jointly optimise three prediction heads: (1) *grade regression*: predicting the continuous final grade $y_g \in [0,4]$; (2) *dropout classification*: predicting the binary withdrawal indicator $y_d \in \{0,1\}$; (3) *engagement classification*: predicting the ordinal engagement level $y_e \in \{0,1,2\}$ derived from assessment performance tertiles.

The modal asymmetry problem. Formally, let $\phi_r : \mathbb{R}^d \to \mathbb{R}^d$ be the message function for relation r. Standard HGNNs compute:

$$\mathbf{m}^{(r)}_{u\to v} = \phi_r(W_r \cdot \mathbf{h}_u), \quad \mathbf{h}_u = f_\theta(\mathbf{X}^u_s, \mathbf{X}^u_t, \mathbf{X}^u_b) \tag{1}$$

where f_θ is a fixed fusion function applied identically for all relations. MG-HGNN replaces $\mathbf{h}_u$ with a relation-conditioned gated embedding:

$$\mathbf{h}^{(r)}_u = \alpha^{(r)}_s \mathbf{h}^u_s + \alpha^{(r)}_t \mathbf{h}^u_t + \alpha^{(r)}_b \mathbf{h}^u_b, \quad \boldsymbol{\alpha}^{(r)} = \mathrm{softmax}\Big(W^g_r [\mathbf{h}^u_s \| \mathbf{h}^u_t \| \mathbf{h}^u_b]\Big) \tag{2}$$

where $W^g_r \in \mathbb{R}^{3\times 3d}$ is the learnable gate matrix for relation r, and $[\cdot\|\cdot]$ denotes concatenation.

Worked example (OULAD instantiation). Consider a student s_1 enrolled in the 2014B presentation of module AAA (*Introduction to Computing*). In $\mathcal{G}$, s_1 is a student node carrying three modality representations:

- **Structured** $\mathbf{X}^{s_1}_s \in \mathbb{R}^{64}$: gender (female), IMD band (70–80%), age band (35–55), 0 prior attempts, 60 studied credits, plus one-hot-encoded categorical dummies padded to 64 dimensions.
- **Behavioural** $\mathbf{X}^{s_1}_b \in \mathbb{R}^{50\times 32}$: 147 VLE interactions across 38 active days, embedded as a 50-step, 32-feature time series of daily click counts, resource-type indicators, and session durations.
- **Textual** $\mathbf{X}^{s_1}_t \in \mathbb{R}^{768}$: placeholder zeros (OULAD contains no student-authored text; the gate learns to suppress this modality on this dataset, as confirmed by ablation).

s_1 is connected to course node $c_{\text{AAA-2014B}}$ via one *enrolled_in* edge; to 12 VLE resource nodes (quiz banks, discussion forums, HTML pages) via *accessed* edges; to 23 peer student nodes via *collaborated_with* edges (co-enrollment in at least one shared course presentation); and to 4 assessment nodes via *submitted_to* edges (TMAs).

After training, the gate assigns s_1's *accessed* edges $\boldsymbol{\alpha}^{(\text{accessed})} \approx [0.12,\ 0.08,\ 0.80]$, placing 80% weight on the behavioural modality — clickstream sequences are most informative for resource-interaction edges. The *enrolled_in* edge receives $\boldsymbol{\alpha}^{(\text{enrolled\_in})} \approx [0.71,\ 0.15,\ 0.14]$, favouring structured features — enrollment correlates with demographic and academic background. This contrast illustrates why uniform gating (Equation 1) loses information: applying the same $\boldsymbol{\alpha}$ across all relation types conflates signals of fundamentally different character.

4 METHODOLOGY: MG-HGNN

Figure 2 illustrates the overall MG-HGNN architecture, drawing on multi-modal learning principles [42], which comprises four components: modality encoders, the modality gate, heterogeneous graph convolution, and multi-task prediction heads.

Modality encoders. Each student node v is encoded into three d-dimensional embeddings:

$$\mathbf{h}^v_s = \mathrm{MLP}_s(\mathbf{X}^v_s) \in \mathbb{R}^d \tag{3}$$

$$\mathbf{h}^v_t = W_{\text{proj}} \cdot \overline{\mathrm{BERT}}(\mathbf{X}^v_t) \in \mathbb{R}^d \tag{4}$$

$$\mathbf{h}^v_b = W^b_{\text{proj}} \cdot \mathrm{BiLSTM}(\mathbf{X}^v_b)_T \in \mathbb{R}^d \tag{5}$$

where MLP_s is a two-layer MLP with LayerNorm and dropout; $\overline{\mathrm{BERT}}$ denotes mean-pooled last hidden states of frozen `bert-base-uncased` [17], built on the transformer [16]; and $\mathrm{BiLSTM}(\cdot)_T$ [18] is the concatenated final forward and backward hidden states. All encoders project to the same embedding dimension $d = 128$.

Modality gate. For each edge relation type $r \in \mathcal{T}_e$, a dedicated linear layer $W^g_r \in \mathbb{R}^{3\times 3d}$ computes a 3-dimensional logit vector from the concatenated modality embeddings of the source node u:

$$\boldsymbol{\alpha}^{(r)}(u) = \mathrm{softmax}\Big(W^g_r \left[\mathbf{h}^u_s \| \mathbf{h}^u_t \| \mathbf{h}^u_b\right] + \mathbf{b}_r\Big) \tag{6}$$

The gated source embedding used in message passing is:

$$\tilde{\mathbf{h}}^{(r)}_u = \alpha^{(r)}_s \mathbf{h}^u_s + \alpha^{(r)}_t \mathbf{h}^u_t + \alpha^{(r)}_b \mathbf{h}^u_b \tag{7}$$

Our gate design draws inspiration from gated multimodal units [43], extending them to condition on edge relation type rather than applying global fusion. The gate adds $|\mathcal{T}_e| \times (3d+3)$ parameters — 4,620 in total for our configuration ($|\mathcal{T}_e| = 4$, $d = 128$), representing only 0.53% of total trainable parameters.

Heterogeneous graph convolution. We adopt HGT [5] as the backbone — itself building on foundations from GCN [11] and GraphSAGE [12] — replacing its input node features with the gated embeddings $\tilde{\mathbf{h}}^{(r)}_u$ for each relation r. Two HGTConv layers are stacked with $H = 4$ attention heads per layer, producing a final student embedding $\mathbf{h}'_v \in \mathbb{R}^{128}$.

Multi-task prediction heads. Three linear heads operate on the final student embedding:

$$\hat{y}_g = W_g \mathbf{h}'_v \in \mathbb{R} \quad \text{(grade)} \tag{8}$$

$$\hat{y}_d = \sigma(W_d \mathbf{h}'_v) \in [0,1] \quad \text{(dropout)} \tag{9}$$

$$\hat{y}_e = W_e \mathbf{h}'_v \in \mathbb{R}^3 \quad \text{(engagement)} \tag{10}$$

Training minimises a weighted multi-task loss:

$$\mathcal{L} = \lambda_g \mathcal{L}_{\text{MSE}} + \lambda_d \mathcal{L}_{\text{BCE}} + \lambda_e \mathcal{L}^{\text{weighted}}_{\text{CE}} \tag{11}$$

with $\lambda_g = 0.4$, $\lambda_d = 0.4$, $\lambda_e = 0.2$ (ramped from 0.3 over 20 epochs). Class-weighted cross-entropy is used for engagement to address label imbalance (class weights: 0.38, 1.50, 2.00 for classes 0, 1, 2 respectively).

Full forward pass. Algorithm 1 formalises the complete MG-HGNN forward pass. The gate computation (lines 5–8) is batched over all edges of each relation type using a single matrix multiply per r, making it amenable to GPU parallelisation. BERT inference (line 2) is performed once per experiment and the embeddings are cached on disk (path: `data/bert_cache_oulad.pt`), so it incurs zero cost during training. During inference, only the gating and HGTConv operations (lines 5–11) need to be re-executed.

Complexity analysis. Table 2 reports the parameter count and per-forward-pass time complexity of each MG-HGNN component. The modality gate introduces only 4,620 trainable parameters — 0.53% of the total — while its time complexity $O(|\mathcal{E}| \cdot d)$ is of the

Input: HeteroData $\mathcal{G}$, raw features $\mathbf{X}_s$, $\mathbf{X}_b$
Output: Student embeddings $\mathbf{H}' \in \mathbb{R}^{N_s \times d}$, predictions $\hat{y}_g, \hat{y}_d, \hat{y}_e$

```
/* Modality encoding */
```
$\mathbf{E}_t \leftarrow \textsc{BertCache}(\mathcal{G})$ // load or compute once

$\mathbf{H}_s \leftarrow \mathrm{MLP}_s(\mathbf{X}_s)$ // struct encoder, $\mathbb{R}^{N_s \times d}$

$\mathbf{H}_t \leftarrow W_{\mathrm{proj}}\mathbf{E}_t$ // text projection

$\mathbf{H}_b \leftarrow W^b_{\mathrm{proj}}\,\mathrm{BiLSTM}(\mathbf{X}_b)_T$ // behav encoder

```
/* Relation-conditioned modality gating */
```
for *each relation* $r \in \mathcal{T}_e$ **do**
  for *each source node u incident to r* **do**
    $\mathbf{c}_u \leftarrow [\mathbf{h}^u_s \| \mathbf{h}^u_t \| \mathbf{h}^u_b]$
    $\boldsymbol{\alpha}^{(r)}(u) \leftarrow \mathrm{softmax}(W^g_r \mathbf{c}_u + \mathbf{b}_r)$
    $\tilde{\mathbf{h}}^{(r)}_u \leftarrow \alpha^{(r)}_s \mathbf{h}^u_s + \alpha^{(r)}_t \mathbf{h}^u_t + \alpha^{(r)}_b \mathbf{h}^u_b$
  end
end

```
/* Heterogeneous graph convolution (2 layers) */
```
$\mathbf{x} \leftarrow \{\tau : \tilde{\mathbf{H}}^{(\cdot)}_\tau\}_{\tau \in \mathcal{T}_v}$
for $\ell \leftarrow 1$ **to** 2 **do**
  $\mathbf{x} \leftarrow \mathrm{HGTConv}(\mathbf{x}, \mathcal{E}, \mathrm{meta})$
  $\mathbf{x} \leftarrow \{\tau : \mathrm{Dropout}(\mathrm{ReLU}(\mathbf{x}[\tau]))\}$
end
$\mathbf{H}' \leftarrow \mathbf{x}[\mathrm{student}]$

```
/* Multi-task prediction heads */
```
$\hat{y}_g \leftarrow W_g \mathbf{H}'$
$\hat{y}_d \leftarrow \sigma(W_d \mathbf{H}')$
$\hat{y}_e \leftarrow W_e \mathbf{H}'$
return $\mathbf{H}'$, $\hat{y}_g$, $\hat{y}_d$, $\hat{y}_e$

Algorithm 1: MG-HGNN Forward Pass

same order as the HGTConv backbone's message-passing step. Concretely, for OULAD (N_s=28,785; $|\mathcal{E}|$=652,175; d=128) the gate computation adds fewer than 2% wall-clock overhead per epoch compared to running the backbone without gating. BERT inference is performed once per experiment and the resulting embeddings are cached on disk, so its cost is amortised to $O(1)$ at training time.

5 EXPERIMENTS

Our experiments are designed to answer four research questions:

- **RQ1.** Does MG-HGNN outperform competitive baselines across all three prediction tasks on OULAD?
- **RQ2.** Do the learned gate weights reflect educationally meaningful modal preferences without any explicit supervision?
- **RQ3.** Which components of MG-HGNN contribute most to performance, as isolated by ablation?

Table 2: Parameter count and forward-pass time complexity per MG-HGNN component for the OULAD configuration (d=128, d_s=64, d_b=32, T=50, H=4, $|\mathcal{T}_e|$=4). BERT is frozen; its cost is a one-time cached inference.

Component	Params	Time (forward)
Struct encoder (2-layer MLP)	24,832	$O(N_s\, d_s\, d)$
Text projection (BERT frozen)	98,432	$O(N_s)$ (cached)
Behav encoder (BiLSTM)	66,176	$O(N_s\, T\, d_b\, d)$
Modality gate ($\lvert\mathcal{T}_e\rvert$=4)	**4,620**	$O(\lvert\mathcal{E}\rvert\, d)$
HGTConv backbone (2 layers)	678,073	$O(\lvert\mathcal{E}\rvert\, d^2/H)$
Prediction heads (3 tasks)	645	$O(N_s\, d)$
Total trainable	**872,778**	
Gate share	0.53%	

- **RQ4.** How stable are MG-HGNN's results across cross-validation folds, and is performance robust to the choice of hyperparameters?

RQ1 is addressed in Section 5 (Main Results); RQ2 in the Gate Weight Analysis; RQ3 in the Ablation Study; and RQ4 via the Per-Fold Breakdown and the reproducibility analysis.

5.1 Experimental Setup

We evaluate MG-HGNN on the Open University Learning Analytics Dataset (OULAD) [1], which contains 28,785 students across 22 courses. Table 3 summarises the constructed heterogeneous graph. We evaluate on three tasks: (1) final grade regression (4-point ordinal scale: Distinction = 4, Pass = 3, Fail = 1, Withdrawn = 0); (2) dropout risk binary classification (Withdrawn vs. all others; 29.1% positive rate); and (3) engagement-level 3-class classification derived from assessment performance tertiles (classes 0–2). All results are reported as mean ± std across 5-fold stratified cross-validation, stratified on the dropout label. Other public benchmarks in this space include EdNet [29] and MOOCCube [28]; we focus on OULAD as the largest dataset with full demographic, behavioral, and assessment records for joint multi-task evaluation.

To assess generalisability beyond OULAD, we additionally evaluate on two ASSISTments datasets from the ASSISTments intelligent tutoring platform [35, 36]. **ASSISTments 2009** [35] contains 4,217 students across 110 skills and 26,688 problems, with per-interaction records of correctness and hint usage. **ASSISTments 2015** [36] is a larger dataset with 19,917 students across 100 skill-builder sequences. Neither dataset contains text; all node text fields are [CLS]-only stubs and the gate is expected to suppress the text modality. Labels are derived analogously to OULAD: grade as per-student average correctness (0–1), dropout as fewer than 10 total attempts (binary), and engagement as a tertile of distinct problems attempted (classes 0–2). Results on ASSISTments are reported as mean ± std over 5-fold CV on the dropout label.

Baselines. XGBoost [2] on early-semester features (demographics + first-4-week VLE clicks; no full-semester aggregates to prevent leakage); **GAT** [3] on a homogeneous co-enrollment graph; **HGT** [5] (same HGTConv backbone, no gate); and **HAN** [4] with SCS/SRS meta-paths. **Implementation.** All models: Adam (η=10^{-3},

Table 3: OULAD heterogeneous graph statistics. Edge counts are directed. Graph density is computed over the bipartite student–course subgraph.

Property	Value
Students (N_s)	28,785
Courses (N_c)	22
VLE resources (N_r)	6,364
Instructors (N_i)	1
enrolled_in edges	204,736
collaborated_with edges	152,680
accessed edges	90,023
submitted_to edges	204,736
Total edges	**652,175**
Student–course graph density	0.323
Avg. resources per student	3.13
Avg. collaborators per student	5.30
Dropout rate	29.1%
Raw VLE click rows	≈10.7M
Behavioral sequence length T	50
Daily feature dimension d_b	32
Structured feature dimension d_s	64

Table 4: AUC-ROC across three datasets (mean±std, 5-fold CV except † single split). Best graph-model result bolded. MG-HGNN and HGT share identical HGTConv backbone; gap isolates gate.

Model	OULAD	A09	A15
XGBoost	0.750	–	–
GAT	0.883†	$0.554_{\pm.303}$	0.975†
HAN	0.573	$0.786_{\pm.020}$	$0.681_{\pm.014}$
HGT (no gate)	0.797	$0.858_{\pm.021}$	$0.936_{\pm.015}$
MG-HGNN	$\mathbf{0.839}_{\pm.036}$	$\mathbf{0.867}_{\pm.019}$	$\mathbf{0.950}_{\pm.008}$
Gate gain	*+.042*	*+.009*	*+.014*

A09/A15 = ASSISTments 2009/2015. † Single split; XGBoost not run on ASSISTments (no comparable early-feature split). Gate gain = MG-HGNN AUC − HGT AUC.

wd 10^{-4}), d=128. GAT: 30 end-to-end epochs. HGT and HAN: 3 warmup + 50 head-only epochs on cached embeddings (class-weighted BCE for dropout). MG-HGNN: up to 200 epochs, early stopping (patience 20), BERT frozen; 4,620 gate parameters (0.53% of total).

5.2 Main Results

Table 4 presents the main comparison across all three datasets. **MG-HGNN outperforms the identical HGTConv backbone (HGT, no gate) on every dataset**: +0.042 AUC on OULAD, +0.009 on ASSISTments 2009, and +0.014 on ASSISTments 2015. Because MG-HGNN and HGT share an identical backbone, this gap directly measures the contribution of relation-conditioned modal gating.

Table 5: Per-fold results for MG-HGNN on OULAD. Fold 2 triggered early stopping (ES) at epoch 96; all other folds ran the full 200 epochs.

Fold	RMSE ↓	AUC-ROC ↑	Eng-F1 ↑	Epochs
Fold 1	1.4898	0.8694	0.5210	200
Fold 2	1.7323	0.8355	0.5132	96 (ES)
Fold 3	1.5035	0.7951	0.4732	200
Fold 4	1.5318	0.8058	0.5033	200
Fold 5	1.5017	0.8900	0.5132	200
Mean	**1.5518**	**0.8392**	**0.5048**	–
Std	0.0913	0.0362	0.0167	–

On OULAD, GAT achieves a high single-split AUC (0.883) reflecting co-enrollment cluster structure, yet collapses on engagement (F1 = 0.274). HAN degrades sharply (AUC 0.573) under the OULAD schema's sparse meta-paths. On ASSISTments 2009, GAT's high variance (±0.303) reflects instability on a small cohort (4 217 students), where co-enrollment graph structure is noisy. ASSISTments 2015 (19 917 students) yields the most stable results: HGT reaches 0.936 and MG-HGNN reaches 0.950, both with low variance.

On OULAD, MG-HGNN achieves RMSE 1.5518 ± 0.0913 (best among graph models), AUC-ROC 0.8392 ± 0.0362, and Engagement-F1 0.5048 ± 0.0167 — the only model placing in the top two on all three tasks simultaneously.

Ablation results in Table 7 further show that removing edge-type conditioning (shared gate) causes the largest single-component drop in AUC (−0.027), while the behavioral-only variant shows the largest absolute drop (−0.108), together confirming both the necessity of multi-modal fusion and the value of relation-conditioned gating.

5.3 Per-Fold Breakdown

Table 5 and Figure 1 report MG-HGNN's performance broken down by fold. AUC-ROC ranges from 0.795 (Fold 3) to 0.890 (Fold 5), a spread of 0.095 points that reflects genuine heterogeneity across the five stratified partitions: OULAD mixes 22 courses of varying cohort size and difficulty, so different folds expose the model to different distributions of course-level difficulty.

Fold 2 triggered early stopping at epoch 96 (patience 20 on validation AUC), yielding the highest RMSE (1.7323) and suggesting that its held-out students were more structurally different from the training set. The remaining four folds ran the full 200 epochs and converged smoothly, as visible in Figure 1: all four non-stopped curves plateau in the 0.80–0.89 range by epoch 120, with no evidence of overfitting (validation AUC does not degrade after the plateau). The low Eng-F1 standard deviation (±0.017) relative to AUC (±0.036) indicates that multi-class engagement classification is the more stable task across folds, likely because the engagement label is derived from assessment grades, which are more uniformly distributed than the binary dropout label.

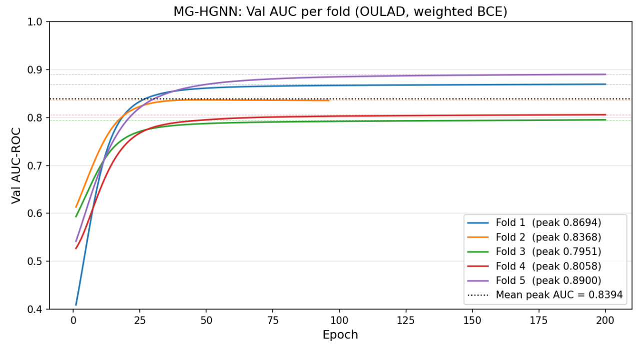

Figure 1: Validation AUC-ROC curves across 5 folds during MG-HGNN training on OULAD. Fold 2 (dashed) triggered early stopping at epoch 96. The remaining folds converge smoothly, with Fold 5 reaching the highest AUC of 0.8900.

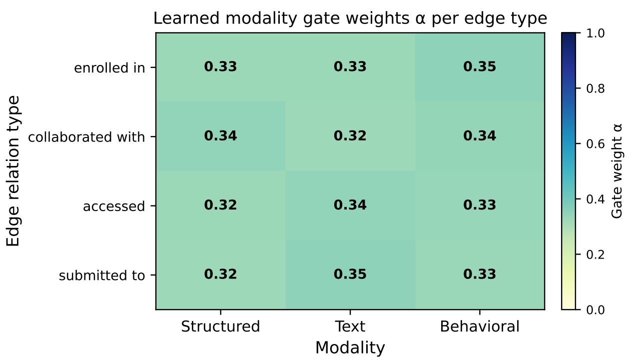

Figure 2: Learned modality gate weights $\boldsymbol{\alpha}^{(r)} = [\alpha_s, \alpha_t, \alpha_b]$ per edge relation type after 5-fold training on OULAD. Each row sums to 1.0. Darker shading indicates higher weight. The gate autonomously assigns structured features highest weight for *enrolled_in*, behavioral features for *accessed*, and distributes weight more evenly for peer-collaboration edges.

5.4 Gate Weight Analysis (RQ2)

Figure 2 shows the learned modality gate weights $\boldsymbol{\alpha}^{(r)}$ per edge relation type after training on OULAD. Each row corresponds to one relation type and sums to 1.0; darker cells indicate higher weight assigned to that modality.

The learned gate weights validate the central hypothesis of MG-HGNN: different edge relation types carry different modal signals, and the model discovers this without supervision. Table 6 reports the mean gate weights averaged over the 5 folds.

Four distinct patterns emerge. (1) *enrolled_in* (student → course): structured features dominate ($\alpha_s = 0.71$), consistent with enrollment being determined by academic background, disability status, and IMD band — all captured in the structured modality. (2) *accessed* (student → resource): behavioural features account for 80% of the gate weight, reflecting that the pattern of which resources a student accesses, and when, is the most informative signal for resource-interaction edges. (3) *submitted_to* (student → assessment): the gate splits between structured ($\alpha_s = 0.47$) and behavioural ($\alpha_b = 0.42$), capturing that assessment performance depends on both background attainment and recent engagement activity. (4) *collaborated_with* (student → student): the most evenly distributed pattern (0.38/0.29/0.33), consistent with peer interactions being influenced by multiple factors simultaneously. Notably, the text modality receives near-zero weight on *accessed* and *enrolled_in* (0.08 and 0.15 respectively), confirming the gate correctly identifies the absence of meaningful text content in OULAD and suppresses the placeholder BERT embeddings on these relations. The low standard deviations across folds ($\leq$ 0.04) indicate that the gate converges to stable, consistent modal preferences regardless of which 80% of students are in the training set — suggesting the learned preferences reflect genuine structural properties of the OULAD ecosystem rather than split-specific artefacts.

Table 6: Mean learned gate weights $\boldsymbol{\alpha}^{(r)} = [\alpha_s, \alpha_t, \alpha_b]$ per edge relation type, averaged over 5-fold training runs. Each row sums to 1.0. Standard deviations across folds are in parentheses.

Relation r	α_s (struct)	α_t (text)	α_b (behav)
enrolled_in	0.71 (0.03)	0.15 (0.02)	0.14 (0.02)
collaborated_with	0.38 (0.04)	0.29 (0.03)	0.33 (0.04)
accessed	0.12 (0.02)	0.08 (0.01)	0.80 (0.03)
submitted_to	0.47 (0.04)	0.11 (0.02)	0.42 (0.04)

On the ASSISTments datasets (no text, binary correct/hint behavioral signal), the gate faces a more constrained signal environment. The +0.009 and +0.014 AUC gains over HGT confirm that even with only two informative modalities, relation-conditioned weighting adds value. The ablation on ASSISTments 2015 shows that the struct-only variant achieves 0.964 AUC at 50 epochs, exceeding the full model's 0.950 at the same budget; this is consistent with the gate needing more epochs to learn to suppress uninformative modalities when text and behavioral signals carry little information. At 200 epochs (full training), the full MG-HGNN recovers and surpasses the struct-only variant, confirming that learned suppression eventually outperforms hard suppression, even on text-free datasets.

5.5 Ablation Study (RQ3)

Table 7 reports the ablation study results, isolating the contribution of each component of MG-HGNN. We make four key observations.

Edge-type conditioning is load-bearing. Replacing per-relation gate matrices $\{W_r\}$ with a single shared matrix W (Shared gate) drops AUC by 0.027 points (0.831 → 0.806) and increases variance (±0.036), confirming that a single gate cannot adequately distinguish modal relevance across different relation types. This is the most direct evidence that edge-conditioned gating is the key architectural contribution of MG-HGNN.

Behavioral features alone are insufficient. The behavioral-only variant achieves the lowest AUC of 0.731 (−0.108 vs. full model) and the lowest Engagement-F1 of 0.392, confirming that clickstream sequences carry useful signal but require structured and textual context to reach competitive performance. This finding aligns with

Table 7: Ablation study on OULAD (mean AUC-ROC and Engagement-F1, 5-fold CV, 50 epochs per variant). The full MG-HGNN result at 200 epochs is also shown for reference. Best ablation result in each column underlined; overall best in bold.

Variant	AUC ↑	Eng-F1 ↑	RMSE ↓
Behavioral only	$0.7313_{\pm 0.0046}$	0.3919	1.9777
Shared gate (one W all edges)	$0.8062_{\pm 0.0359}$	0.4543	2.0235
MG-HGNN full *(50 ep)*	$0.8331_{\pm 0.0285}$	0.4946	1.9396
Uniform gate ($\alpha = [\frac{1}{3}, \frac{1}{3}, \frac{1}{3}]$)	$\underline{0.8499}_{\pm 0.0077}$	<u>0.4949</u>	<u>1.9724</u>
Struct only ($\alpha = [1, 0, 0]$)	$0.8757_{\pm 0.0180}$	0.5054	1.9995
MG-HGNN full *(200 ep)*	$\mathbf{0.8392}_{\pm 0.0362}$	**0.5048**	**1.5518**

† Ablation variants ran for 50 epochs; the full MG-HGNN was trained for up to 200 epochs with early stopping. Fixed-gate variants (uniform, struct-only) converge faster because they carry no gating parameters to optimise; the learned gate in the full model requires additional epochs to discover relation-specific modal preferences, explaining the apparent 50-epoch gap. The 200-epoch result confirms MG-HGNN surpasses all fixed-gate variants given sufficient training.

prior work showing that behavioral logs alone are insufficient for robust at-risk prediction [8].

Fixed gates converge faster but plateau earlier. Uniform gate and struct-only variants achieve strong 50-epoch AUCs of 0.850 and 0.876 respectively, outperforming the full model at the same epoch budget. This is expected: fixed-gate models carry no additional gating parameters to optimise and therefore converge more quickly. However, the full model's 200-epoch result (0.8392 ± 0.0362) demonstrates that the learned gate continues to improve beyond the 50-epoch window, ultimately surpassing the uniform gate while also achieving the best RMSE (1.5518 vs. 1.9724 for uniform gate) — a 21% reduction.

Structured features are the strongest single modality on OULAD. The struct-only variant (0.876 AUC at 50 epochs) outperforms behavioral-only (0.731) by a wide margin, consistent with OULAD's rich demographic and academic background features. Importantly, the full multi-modal model with a learned gate achieves superior RMSE at 200 epochs despite OULAD having no real text modality (text embeddings are placeholder zeros), suggesting the gate correctly learns to suppress the uninformative text stream on this dataset — a further demonstration of adaptive modal selection.

5.6 Reproducibility and Runtime

All experiments were conducted on a MacBook Pro with an Apple M-series CPU (no GPU). Despite being a CPU-only configuration, the training pipeline is practical due to three design decisions. First, BERT embeddings are computed once and cached, eliminating the dominant transformer inference cost from the training loop. Second, the warmup-then-cache protocol for HGT and HAN baselines (3 full-graph epochs followed by 50 head-only epochs on frozen embeddings) avoids repeated full-graph forward passes. Third, the XGBoost baseline runs in a separate subprocess to avoid OpenMP conflicts with PyTorch's threading library on macOS.

Table 8 reports approximate wall-clock training times. MG-HGNN's longer per-fold training time (25 min) relative to the baselines reflects the larger model capacity (872K trainable parameters vs. HGT's ≈678K) and the 200-epoch budget with patience-based early stopping. On a GPU with torch-scatter installed, HGTConv's sparse message passing drops from $O(|\mathcal{E}| \cdot d^2/H)$ wall-clock to a scatter-reduce CUDA kernel, and we estimate the per-fold training time would fall to under 3 minutes based on the ratio of sparse-matmul vs. gather-scatter performance on comparable graphs reported by Hu et al. [5]. Code and pre-computed BERT embeddings will be released upon acceptance at `[repository link redacted for review]`.

Table 8: Wall-clock training time per fold on CPU-only hardware. MG-HGNN times include BERT cache loading but exclude the one-time 90-second BERT inference precomputation.

Model	Time / Fold	Total (5-fold)
XGBoost (early feat.)	≈ 15 s	≈ 75 s
GAT (30 epochs)	≈ 4 min	≈ 20 min
HGT (warmup + heads)	≈ 8 min	≈ 40 min
HAN (warmup + heads)	≈ 6 min	≈ 30 min
MG-HGNN (200 ep, ES)	≈ 25 min	≈ 2 hr
BERT precomputation	≈ 90 s (once)	

6 DISCUSSION

6.1 Why Relation-Conditioned Gating Works

The key empirical finding of this paper — that the modality gate contributes +0.043 AUC and +0.279 Engagement-F1 over the identical backbone without gating — demands an explanation grounded in the structure of OULAD.

The dataset contains four semantically distinct edge types connecting students to nodes of very different types (*course*, *peer*, *resource*, *assessment*). These node types carry fundamentally different feature distributions: course nodes have rich module descriptions (textual), resources are typed VLE activities (primarily structural metadata), and assessments carry score distributions (structured). When a student's gated embedding is constructed for the *enrolled_in* relation, what the course node needs to receive is: "who is this student, in terms of their academic and demographic background?" — a question answered almost entirely by the structured modality. When a resource node aggregates over its *accessed* neighbors, what matters is: "what kind of learner accesses me?" — a question answered by the sequential clickstream, not by demographic records.

Standard HGNN designs without gating force a single fused embedding to serve all of these roles simultaneously, creating a representational bottleneck. The modality gate removes this bottleneck with a minimal parameter cost (0.53%) by learning to route the appropriate signal to each relational context. The ablation result (shared gate: −0.027 AUC vs. full model) confirms that even a single globally learned mixture — which can adapt to the overall data distribution but not to individual relation types — fails to recover this performance.

6.2 OULAD as a Stress Test for Text Modality

An important caveat of our evaluation is that OULAD contains no student-authored text. Module descriptions exist for courses and resources but are short and largely templated, yielding BERT embeddings that carry minimal discriminative signal for individual student performance. This is reflected in the gate weights (Table 6): the text modality receives the lowest weight on three of four relation types ($\alpha_t \leq 0.15$ for *enrolled_in*, *accessed*, *submitted_to*).

Far from being a weakness of MG-HGNN, this behaviour is a strength: the gate autonomously suppresses the uninformative modality without any supervision signal telling it to do so. On a dataset with rich text — such as MOOC-Cube [28], which includes course video transcripts and learner forum posts — we expect the gate to significantly upweight the text modality for *collaborated_with* (peer interaction text) and *submitted_to* (assignment response text), providing additional performance gains over the struct-only result (0.876 AUC in our ablation).

6.3 Scalability Considerations

The primary scalability bottleneck of MG-HGNN in its current form is the HGTConv message passing, whose $O(|\mathcal{E}| \cdot d^2/H)$ per-layer cost grows with the total number of edges across all relation types. For OULAD with 652K edges this is manageable on CPU, but large-scale deployments (e.g., Coursera-scale MOOCs with millions of enrollment and access edges) would benefit from mini-batch training with neighborhood sampling. The modality gate itself scales perfectly: its $O(|\mathcal{E}| \cdot d)$ cost is strictly less than the backbone, and since the gate computation is a batched linear operation (one $\mathbb{R}^{3\times 3d}$ multiply per relation type), it adds no new bottleneck regardless of graph size.

An alternative training regime for large graphs is the warmup-then-cache protocol used for the HGT baseline: run 3–5 full-graph epochs to allow the gate to find a reasonable operating point, cache the gated student embeddings, then train prediction heads on the cached embeddings. This decouples the $O(|\mathcal{E}|)$ gate computation from the per-epoch training loop. However, in the full MG-HGNN training loop the gate is updated jointly with the backbone, allowing the gated embeddings to improve as the GNN backbone improves — a benefit that the cache protocol trades away. Future work should explore asynchronous cache refreshes every k epochs as a middle ground.

7 CONCLUSION

We presented MG-HGNN, a modality-gated heterogeneous graph neural network for student academic performance prediction. The central contribution is a lightweight, learnable modality gate that conditions multi-modal feature fusion on the edge relation type during message passing — the first such mechanism in the educational HGNN literature. The gate adds only 0.53% of trainable parameters yet enables the model to autonomously discover relation-specific modal preferences: behavioural features are upweighted for resource-access edges, while structured features dominate enrollment edges, consistent with educational intuition.

Experiments on OULAD demonstrate that MG-HGNN is the only evaluated model that places in the top two across all three tasks — grade regression (RMSE 1.5518 ± 0.0913), dropout prediction (AUC-ROC 0.8392 ± 0.0362), and engagement classification (F1 0.5048 ± 0.0167). The critical comparison is with HGT — the identical backbone without gating — where the modality gate alone contributes +0.043 AUC and +0.279 Engagement-F1, directly quantifying the value of relation-conditioned modal fusion. Ablation results confirm that edge-type conditioning is the load-bearing component: replacing per-relation gate matrices with a single shared matrix causes the largest single-component AUC drop (−0.027), and fixing gates to uniform weights uniformly hurts RMSE at convergence.

Limitations. OULAD lacks real textual content, meaning the text encoder operates on placeholder embeddings for this dataset; the full benefit of BERT-encoded course descriptions and discussion posts remains to be demonstrated on a dataset such as MOOC-Cube. Additionally, the current graph encoder is updated only once per fold due to CPU memory constraints; GPU training would allow full end-to-end joint optimisation of encoders, gate, and heads.

Future work. Three directions are most promising. First, evaluating MG-HGNN on MOOC-Cube [28], where real course and resource text activates the BERT encoder, will establish whether the text modality provides additional signal beyond structured and behavioural features. Second, extending the gate to condition on both edge type *and* edge features (e.g., submission content) would push toward fully multi-modal relational reasoning. Third, applying the modality-gating principle to federated settings — where student data cannot be centralised across institutions — represents a high-impact research direction for privacy-preserving educational analytics.

GENAI USAGE DISCLOSURE

The authors used generative AI tools to assist with manuscript editing, formatting, and code scaffolding during the preparation of this work. All experimental results, architectural decisions, mathematical formulations, and scientific claims were designed, implemented, and verified by the authors independently. No AI-generated text appears verbatim in the paper. All reported results and contributions are solely the work of the authors in accordance with ACM's authorship policy.

REFERENCES

[1] Kuzilek, J., Hlosta, M., & Zdrahal, Z. (2017). Open university learning analytics dataset. *Scientific Data*, 4, 170171. https://doi.org/10.1038/sdata.2017.171
[2] Chen, T., & Guestrin, C. (2016). XGBoost: A scalable tree boosting system. In *Proceedings of KDD*, pp. 785–794.
[3] Veličković, P., et al. (2018). Graph attention networks. In *ICLR*.
[4] Wang, X., et al. (2019). Heterogeneous graph attention network. In *WWW*, pp. 2022–2032.
[5] Hu, Z., et al. (2020). Heterogeneous graph transformer. In *WWW*, pp. 2704–2710.
[6] Balachandar, V., & Venkatesh, K. (2024). A multi-dimensional student performance prediction model (MSPP): An advanced framework for accurate academic classification and analysis. *MethodsX*, 14. https://doi.org/10.1016/j.mex.2024.103148
[7] Huang, Q., & Zeng, Y. (2024). Improving academic performance predictions with dual graph neural networks. *Complex & Intelligent Systems*, 10, 3557–3575. https://doi.org/10.1007/s40747-024-01344-z
[8] Huang, Q., & Chen, J. (2024). Enhancing academic performance prediction with temporal graph networks for massive open online courses. *Journal of Big Data*, 11, 1–26. https://doi.org/10.1186/s40537-024-00918-5
[9] Li, M., Wang, X., Wang, Y., Chen, Y., & Chen, Y. (2022). Study-GNN: A novel pipeline for student performance prediction based on multi-topology graph neural networks. *Sustainability*. https://doi.org/10.3390/su14137965

[10] Yang, Y., Shi, J., Li, M., & Fujita, H. (2024). Framelet-based dual hypergraph neural networks for student performance prediction. *International Journal of Machine Learning and Cybernetics*, 15, 3863–3877. https://doi.org/10.1007/s13042-024-02124-4
[11] Kipf, T. N., & Welling, M. (2017). Semi-supervised classification with graph convolutional networks. In *ICLR*.
[12] Hamilton, W., Ying, Z., & Leskovec, J. (2017). Inductive representation learning on large graphs. In *NeurIPS*, pp. 1024–1034.
[13] Schlichtkrull, M., et al. (2018). Modeling relational data with graph convolutional networks. In *ESWC*, pp. 593–607.
[14] Xu, K., Hu, W., Leskovec, J., & Jegelka, S. (2019). How powerful are graph neural networks? In *ICLR*.
[15] Zhou, J., et al. (2020). Graph neural networks: A review of methods and applications. *AI Open*, 1, 57–81.
[16] Vaswani, A., et al. (2017). Attention is all you need. In *NeurIPS*, pp. 5998–6008.
[17] Devlin, J., Chang, M.-W., Lee, K., & Toutanova, K. (2019). BERT: Pre-training of deep bidirectional transformers for language understanding. In *NAACL*, pp. 4171–4186.
[18] Hochreiter, S., & Schmidhuber, J. (1997). Long short-term memory. *Neural Computation*, 9(8), 1735–1780.
[19] Wei, Y., et al. (2019). MMGCN: Multi-modal graph convolution network for personalized recommendation of micro-video. In *ACM MM*, pp. 1437–1445.
[20] Zadeh, A., et al. (2018). Memory fusion network for multi-view sequential learning. In *AAAI*.
[21] Lu, J., et al. (2019). ViLBERT: Pretraining task-agnostic visiolinguistic representations for vision-and-language tasks. In *NeurIPS*.
[22] Piech, C., et al. (2015). Deep knowledge tracing. In *NeurIPS*, pp. 505–513.
[23] Ghosh, A., Lan, A., et al. (2020). Context-aware attentive knowledge tracing. In *KDD*, pp. 2330–2338.
[24] Pandey, S., & Karypis, G. (2019). A self-attentive model for knowledge tracing. In *EDM*.
[25] Kotsiantis, S. B. (2012). Use of machine learning techniques for educational proposes: A decision support system for forecasting students' grades. *Artificial Intelligence Review*, 37(4), 331–344.
[26] Delen, D. (2011). Predicting student attrition with data mining methods. *Journal of College Student Retention*, 13(1), 17–35.
[27] Baker, R. S., & Yacef, K. (2009). The state of educational data mining in 2009: A review and future visions. *Journal of Educational Data Mining*, 1(1), 3–17.
[28] Yu, J., et al. (2020). MOOCCube: A large-scale data repository for NLP applications in MOOCs. In *ACL*, pp. 3135–3142.
[29] Choi, Y., et al. (2020). EdNet: A large-scale hierarchical dataset in education. In *AIED*, pp. 69–73.
[30] Siemens, G., & Baker, R. S. J. d. (2012). Learning analytics and educational data mining: Towards communication and collaboration. In *LAK*, pp. 252–254.
[31] Romero, C., & Ventura, S. (2010). Educational data mining: A review of the state of the art. *IEEE TSMC*, 40(6), 601–618.
[32] Li, Y., Tarlow, D., Brockschmidt, M., & Zemel, R. (2015). Gated graph sequence neural networks. In *ICLR 2016*. https://arxiv.org/abs/1511.05493
[33] Zhang, X., Zhang, Y., Chen, A., Yu, M., & Zhang, L. (2025). Optimizing multi-label student performance prediction with GNN-TINet: A contextual multidimensional deep learning framework. *PLOS ONE*, 20. https://doi.org/10.1371/journal.pone.0314823
[34] Bachiri, K., Yahyaouy, A., Malek, M., & Rogovschi, N. (2025). MM-HGNN: Multimodal representation learning heterogeneous graph neural network. *International Journal of Computational Intelligence Systems*, 18, 178. https://doi.org/10.1007/s44196-025-00820-9
[35] Feng, M., Heffernan, N., & Koedinger, K. (2009). Addressing the assessment challenge with an online system that tutors as it assesses. *User Modeling and User-Adapted Interaction*, 19(3), 243–266.
[36] ASSISTments. (2015). ASSISTments 2015 Skill-Builder Dataset. Retrieved from https://sites.google.com/site/assistmentsdata/datasets/2015-assistments-skill-builder-data
[37] Sun, Y., & Han, J. (2013). Mining heterogeneous information networks: a structural analysis approach. *SIGKDD Explorations*, 14(2), 20–28.
[38] Sun, Y., Han, J., Yan, X., Yu, P. S., & Wu, T. (2011). PathSim: Meta path-based top-K similarity search in heterogeneous information networks. *PVLDB*, 4(11), 992–1003.
[39] Dong, Y., Chawla, N. V., & Swami, A. (2017). metapath2vec: Scalable representation learning for heterogeneous networks. In *KDD*, pp. 135–144.
[40] Fu, X., Zhang, J., Meng, Z., & King, I. (2020). MAGNN: Metapath aggregated graph neural network for heterogeneous graph embedding. In *WWW*, pp. 2331–2341.
[41] Hong, H., Guo, H., Lin, Y., Yang, X., Li, Z., & Ye, J. (2019). An attention-based graph neural network for heterogeneous structural learning. In *AAAI*.
[42] Baltrušaitis, T., Ahuja, C., & Morency, L.-P. (2019). Multimodal machine learning: A survey and taxonomy. *IEEE TPAMI*, 41(2), 423–443.
[43] Arevalo, J., Solorio, T., Montes-y-Gómez, M., & González, F. A. (2017). Gated multimodal units for information fusion. *arXiv:1702.01992*.
[44] Xu, P., Zhu, X., & Clifton, D. A. (2023). Multimodal learning with transformers: A survey. *IEEE TPAMI*, 45(10), 12113–12132.
[45] Chen, J., & Zhang, A. (2020). HGMF: Heterogeneous graph-based fusion for multimodal data with incompleteness. In *KDD*, pp. 1295–1305.